Medtronic

53401

Single Chamber Temporary External Pacemaker

Electromagnetic Compatibility Declaration

1 Electromagnetic compatibility declaration

The Medtronic Model 53401 Single Chamber Temporary External Pacemaker (hereafter referred to as the “temporary pacemaker”) is tested with the Medtronic 5433A and 5433V Patient Cables, and meets the requirements of IEC 60601-1-2:2007/AC:2010.

- Use of accessories and cables other than those that are specifically listed may result in increased emissions or decreased immunity of the temporary pacemaker.
- The temporary pacemaker should not be used adjacent to or stacked with other equipment and that if adjacent or stacked use is necessary, the temporary pacemaker should be observed to verify normal operation in the configuration in which it will be used.

Guidance and manufacturer’s declaration—electromagnetic emissions		
The temporary pacemaker is intended for use in the electromagnetic environment specified below. The customer or the user of the temporary pacemaker should assure that it is used in such an environment.		
Emissions test	**Compliance**	**Electromagnetic environment—guidance**
RF emissions CISPR 11	Group 1	The temporary pacemaker uses RF energy primarily for its internal function. Therefore, its RF emissions are very low and are not likely to cause interference in nearby electronic equipment.
RF emissions CISPR 11	Class A	The temporary pacemaker is suitable for use in all establishments other than domestic and those directly connected to the public low-voltage power supply network that supplies buildings used for domestic purposes.
Harmonic emissions IEC 61000-3-2	Not Applicable	
Voltage fluctuations/flicker emissions IEC 61000-3-3	Not Applicable	

Guidance and manufacturer’s declaration—electromagnetic immunity			
The temporary pacemaker is intended for use in the electromagnetic environment specified below. The customer or the user of the temporary pacemaker should assure that it is used in such an environment.			
Immunity test	**IEC 60601 Test level**	**Compliance level**	**Electromagnetic environment—guidance**
Electrostatic discharge (ESD) IEC 61000-4-2	±6 kV contact ±8 kV air	±6 kV contact ±8 kV air	Floors should be wood, concrete or ceramic tile. If floors are covered with synthetic material, the relative humidity should be at least 30%.
Electrical fast transient/burst IEC 61000-4-4	Not applicable	Not applicable	
Surge IEC 61000-4-5	Not applicable	Not applicable	
Voltage dips, short interruptions and voltage variations on power supply input lines IEC 61000-4-11	Not applicable	Not applicable	

Guidance and manufacturer’s declaration—electromagnetic immunity			
Power frequency (50/60 Hz) magnetic field IEC 61000-4-8	3 A/m	3 A/m	Power frequency magnetic fields should be at levels characteristic of a typical location in a typical commercial or hospital environment.

Guidance and manufacturer’s declaration—electromagnetic immunity				
The temporary pacemaker is intended for use in the electromagnetic environment specified below. The customer or the user of the temporary pacemaker should assure that it is used in such an environment.				
Immunity test	**IEC 60601 Test level**	**Compliance level**	**Electromagnetic environment—guidance**	
			Portable and mobile RF communications equipment should be used no closer to any part of the temporary pacemaker, including cables, than the recommended separation distance calculated from the equation applicable to the frequency of the transmitter. **Recommended separation distance**	
Conducted RF	3 V_{RMS} 150 kHz to 80 MHz outside ISM bands[a]	10 V	$d = 0.35\sqrt{P}$	
IEC 61000-4-6	10 V_{RMS} V_{RMS} 150 kHz to 80 MHz outside ISM bands[a]	10 V	$d = 1.2\sqrt{P}$	
Radiated RF IEC 61000-4-3	10 V/m 80 MHz to 2.5 GHz	10 V/m	$d = 1.2\sqrt{P}$	80 MHz to 800 MHz
			$d = 2.3\sqrt{P}$	800 MHz to 2.5 GHz
			Where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer and d is the recommended separation distance in meters (m).[b] Field strengths from fixed RF transmitters, as determined by an electromagnetic site survey,[c] should be less than the compliance level in each frequency range.[d] Interference may occur in the vicinity of equipment marked with the following symbol: 	

Guidance and manufacturer's declaration—electromagnetic immunity
Note 1: At 80 MHz and 800 MHz, the higher frequency range applies.
Note 2: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.

[a] The ISM (industrial, scientific, and medical) bands between 150 kHz and 80 MHz are 6.765 MHz to 6.795 MHz; 13.553 MHz to 13.567 MHz; 26.957 MHz to 27.283 MHz; and 40.66 MHz to 40.70 MHz.

[b] The compliance levels in the ISM frequency bands between 150 kHz and 80 MHz and in the frequency range 80 MHz to 2.5 GHz are intended to decrease the likelihood that mobile/portable communications equipment could cause interference if it is inadvertently brought into patient areas. For this reason, an additional factor of 10/3 has been incorporated into the formulae used in calculating the recommended separation distance for transmitters in these frequency ranges.

[c] Field strengths from fixed transmitters, such as base stations for radio (cellular/cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast and TV broadcast cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location in which the temporary pacemaker is used exceeds the applicable RF compliance level above, the temporary pacemaker should be observed to verify normal operation. If abnormal performance is observed, additional measures may be necessary, such as re-orienting or relocating the temporary pacemaker.

[d] Over the frequency range of 150 kHz to 80 MHz, field strengths should be less than 3 V/m.

Recommended separation distances between portable and mobile RF communications equipment and the temporary pacemaker

The temporary pacemaker is intended for use in an electromagnetic environment in which radiated RF disturbances are controlled. The customer or the user of the temporary pacemaker can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communications equipment (transmitters) and the temporary pacemaker as recommended below, according to the maximum output power of the communications equipment.

Rated maximum output power of transmitter	Separation distance according to frequency of transmitter			
	150 kHz to 80 MHz outside ISM bands $d = 0.35\sqrt{P}$	**150 kHz to 80 MHz in ISM bands** $d = 1.2\sqrt{P}$	**80 MHz to 800 MHz** $d = 1.2\sqrt{P}$	**800 MHz to 2.5 GHz** $d = 2.3\sqrt{P}$
0.01 W	0.04 m	0.12 m	0.12 m	0.23 m
0.1 W	0.11 m	0.38 m	0.38 m	0.73 m
1 W	0.35 m	1.2 m	1.2 m	2.3 m
10 W	1.11 m	3.79 m	3.79 m	7.27 m
100 W	3.5 m	12 m	12 m	23 m

For transmitters rated at a maximum output power not listed above, the recommended separation distance *d* in meters (m) can be estimated using the equation applicable to the frequency of the transmitter, where *P* is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer.

Note 1: At 80 MHz and 800 MHz, the separation distance for the higher frequency range applies.

Note 2: The ISM (industrial, scientific, and medical) bands between 150 kHz and 80 MHz are 6.765 MHz to 6.795 MHz; 13.553 MHz to 13.567 MHz; 26.957 MHz to 27.283 MHz; and 40.66 MHz to 40.70 MHz.

Recommended separation distances between portable and mobile RF communications equipment and the temporary pacemaker
Note 3: An additional factor of 10/3 has been incorporated into the formulae used in calculating the recommended separation distance for transmitters in the ISM frequency bands between 150 kHz and 80 MHz and in the frequency range 80 MHz to 2.5 GHz to decrease the liklihood that mobile/portable communications equipment could cause interference if it is inadvertently brought into patient areas.
Note 4: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.

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